

Scenario 1: Student Academic Performance Dashboard

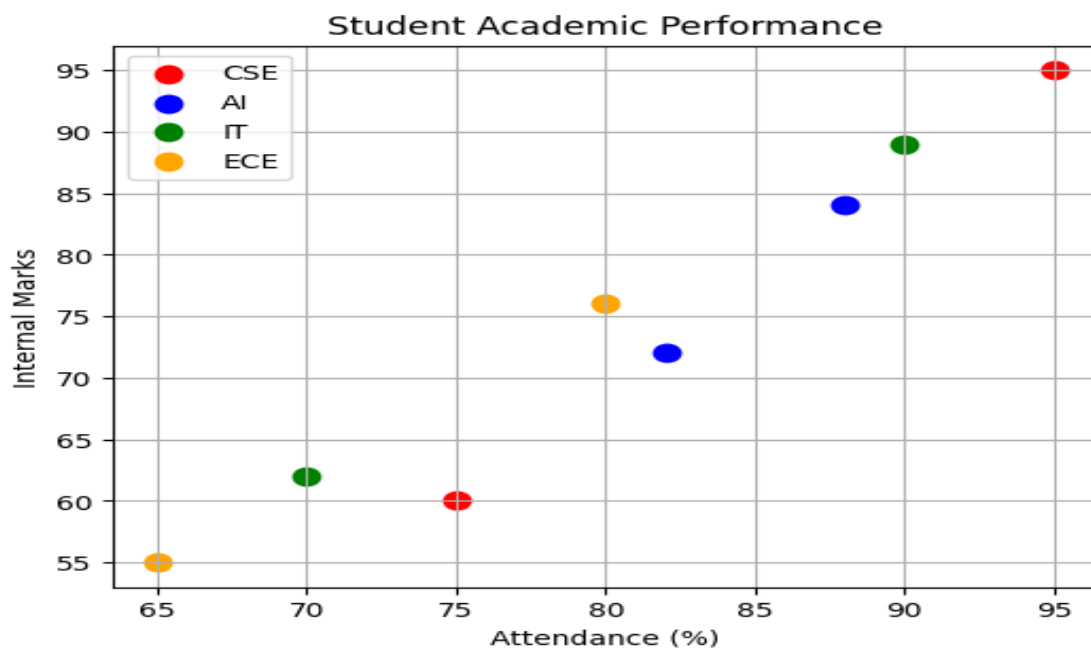
CODE:

```
import pandas as pd
import matplotlib.pyplot as plt

data = {
    'Attendance': [75, 82, 90, 65, 95, 88, 70, 80],
    'Marks': [60, 72, 89, 55, 95, 84, 62, 76],
    'Department': ['CSE', 'AI', 'IT', 'ECE', 'CSE', 'AI', 'IT', 'ECE']
}
df = pd.DataFrame(data)
colors = {'CSE': 'red', 'AI': 'blue', 'IT': 'green', 'ECE': 'orange'}
plt.figure(figsize=(6, 5))

for dept in df['Department'].unique():
    d = df[df['Department'] == dept]
    plt.scatter(d['Attendance'], d['Marks'],
                color=colors[dept], label=dept, s=80)
plt.title("Student Academic Performance")
plt.xlabel("Attendance (%)")
plt.ylabel("Internal Marks")
plt.legend()
plt.grid(True)
plt.show()
```

OUTPUT:

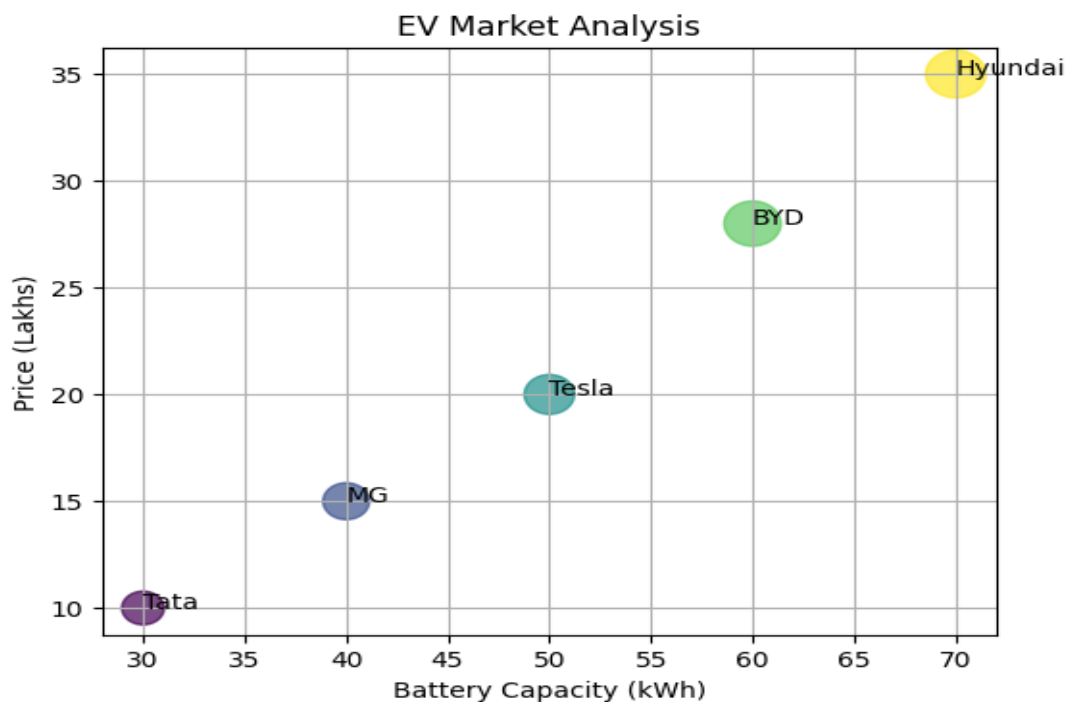


Scenario 2: Electric Vehicle Market Analysis

CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
data = {
    'Battery':[30,40,50,60,70],
    'Price':[10,15,20,28,35],
    'Range':[250,300,350,450,500],
    'Manufacturer':['Tata','MG','Tesla','BYD','Hyundai']
}
df = pd.DataFrame(data)
plt.figure(figsize=(6,5))
plt.scatter(df['Battery'],
            df['Price'],
            s=df['Range'],
            c=range(len(df)),
            alpha=0.7)
for i in range(len(df)):
    plt.text(df['Battery'][i], df['Price'][i], df['Manufacturer'][i])
plt.xlabel("Battery Capacity (kWh)")
plt.ylabel("Price (Lakhs)")
plt.title("EV Market Analysis")
plt.grid(True)
plt.show()
```

OUTPUT:



Scenario 3: Hospital Patient Admission Analysis

CODE:

```
import matplotlib.pyplot as plt

departments = ['Cardiology', 'Neurology', 'Orthopedic', 'General']
patients = [120, 80, 100, 150]
cost = [5000, 7000, 6000, 4000]

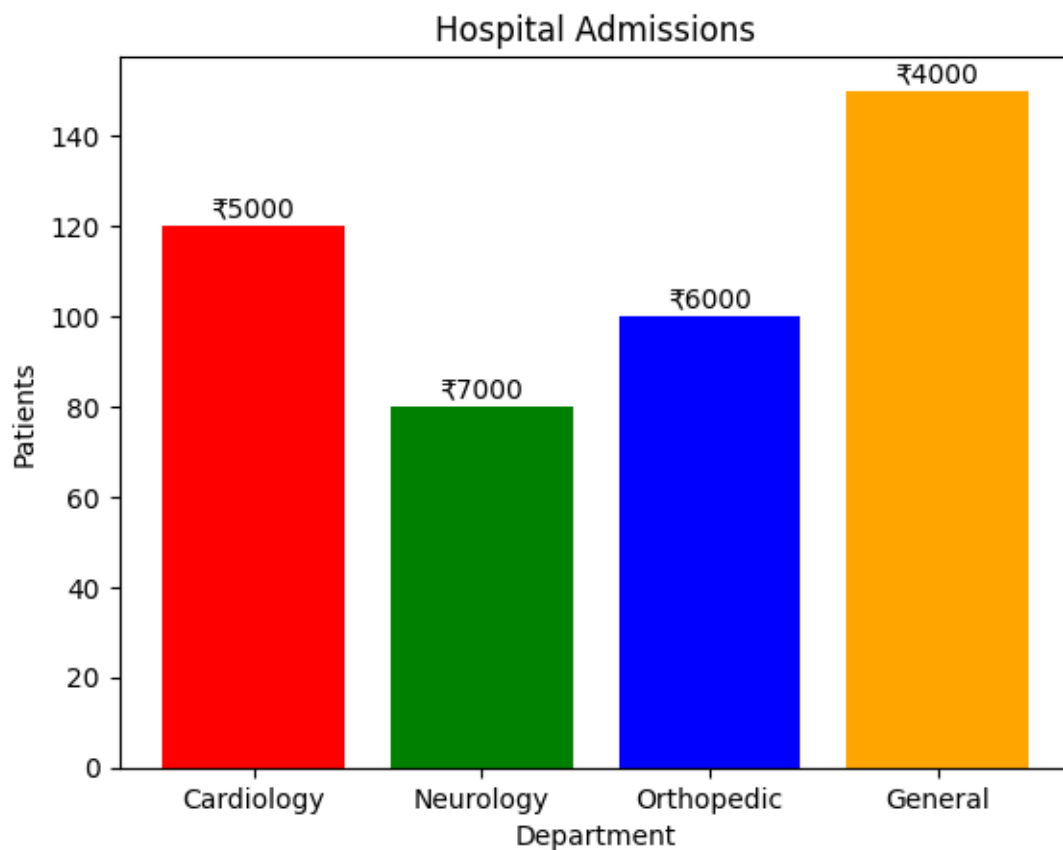
colors = ['red', 'green', 'blue', 'orange']

bars = plt.bar(departments, patients, color=colors)

for i in range(len(bars)):
    plt.text(i, patients[i]+2, "₹"+str(cost[i]), ha='center')

plt.title("Hospital Admissions")
plt.xlabel("Department")
plt.ylabel("Patients")
plt.show()
```

OUTPUT:



Scenario 4: Weather Monitoring System

CODE:

```
import matplotlib.pyplot as plt

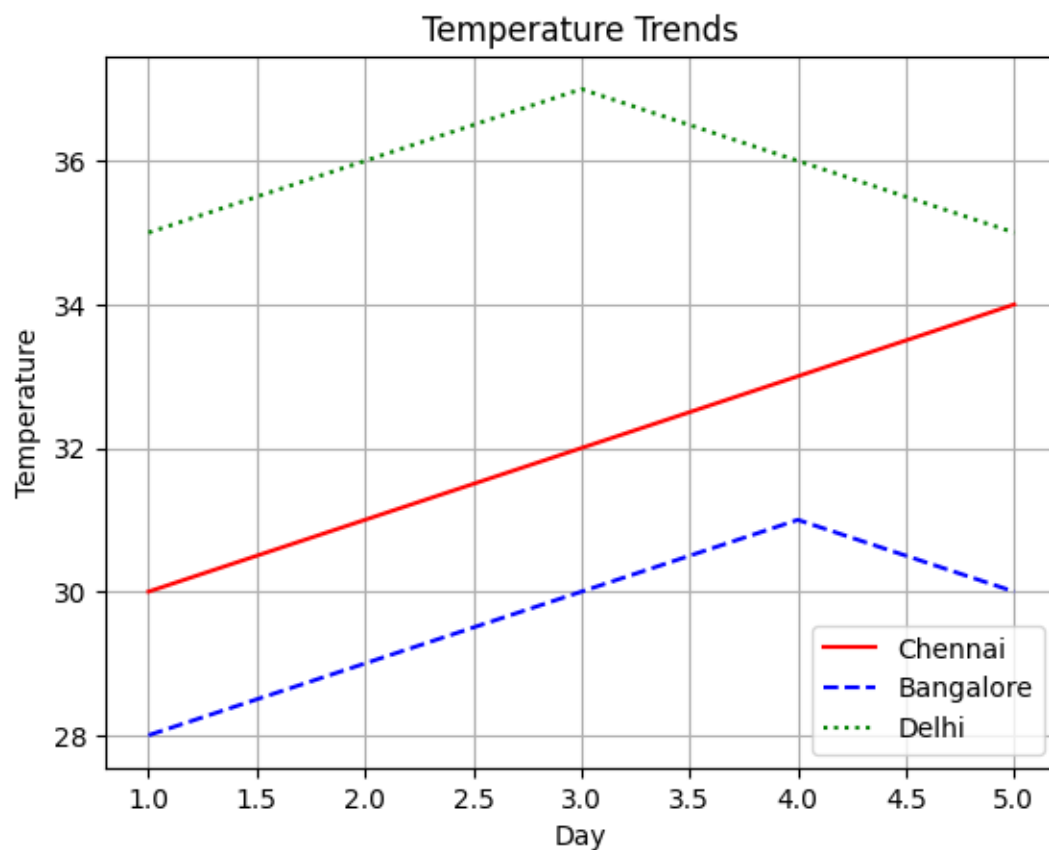
days = [1,2,3,4,5]

city1 = [30,31,32,33,34]
city2 = [28,29,30,31,30]
city3 = [35,36,37,36,35]

plt.plot(days, city1, color='red', linestyle='-', label='Chennai')
plt.plot(days, city2, color='blue', linestyle='--', label='Bangalore')
plt.plot(days, city3, color='green', linestyle=':', label='Delhi')

plt.xlabel("Day")
plt.ylabel("Temperature")
plt.title("Temperature Trends")
plt.legend()
plt.grid(True)
plt.show()
```

OUTPUT:



Scenario 5: Supermarket Sales Analysis

```
import matplotlib.pyplot as plt
import numpy as np

categories = ['Food', 'Drinks', 'Snacks']
branchA = [500, 600, 700]
branchB = [450, 650, 750]

x = np.arange(len(categories))
width = 0.35

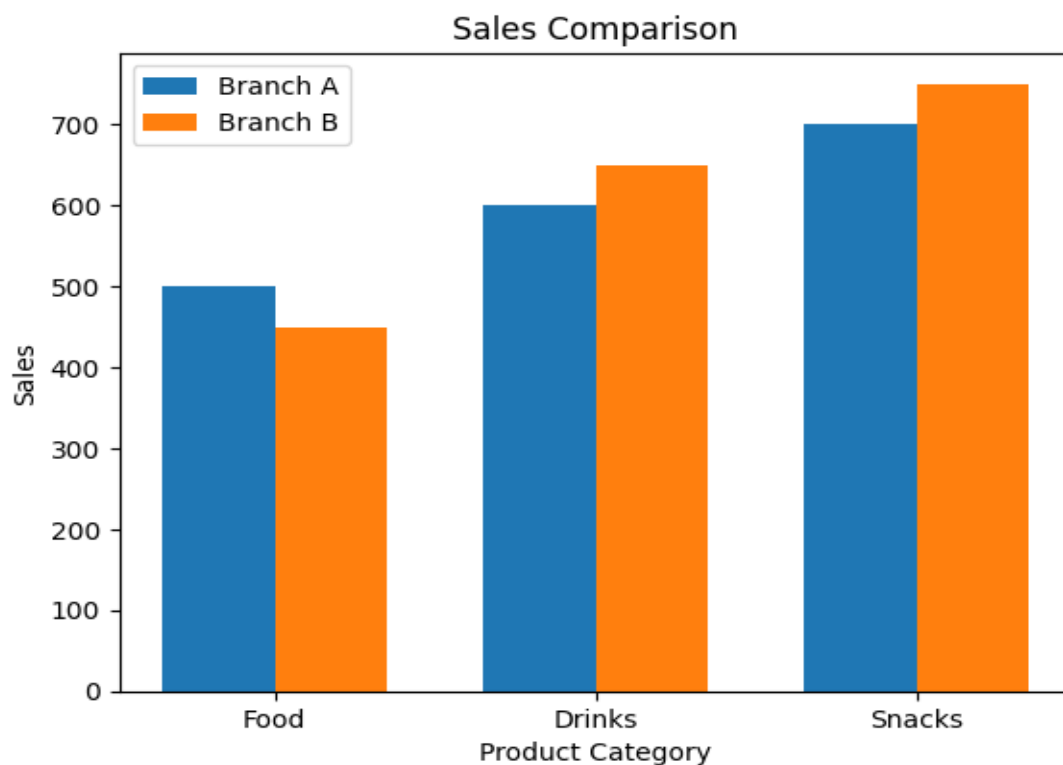
plt.bar(x-width/2, branchA, width, label='Branch A')
plt.bar(x+width/2, branchB, width, label='Branch B')

plt.xticks(x, categories)

plt.xlabel("Product Category")
plt.ylabel("Sales")
plt.title("Sales Comparison")
plt.legend()

plt.show()
```

OUTPUT:



Scenario 6: Employee Salary Distribution

CODE:

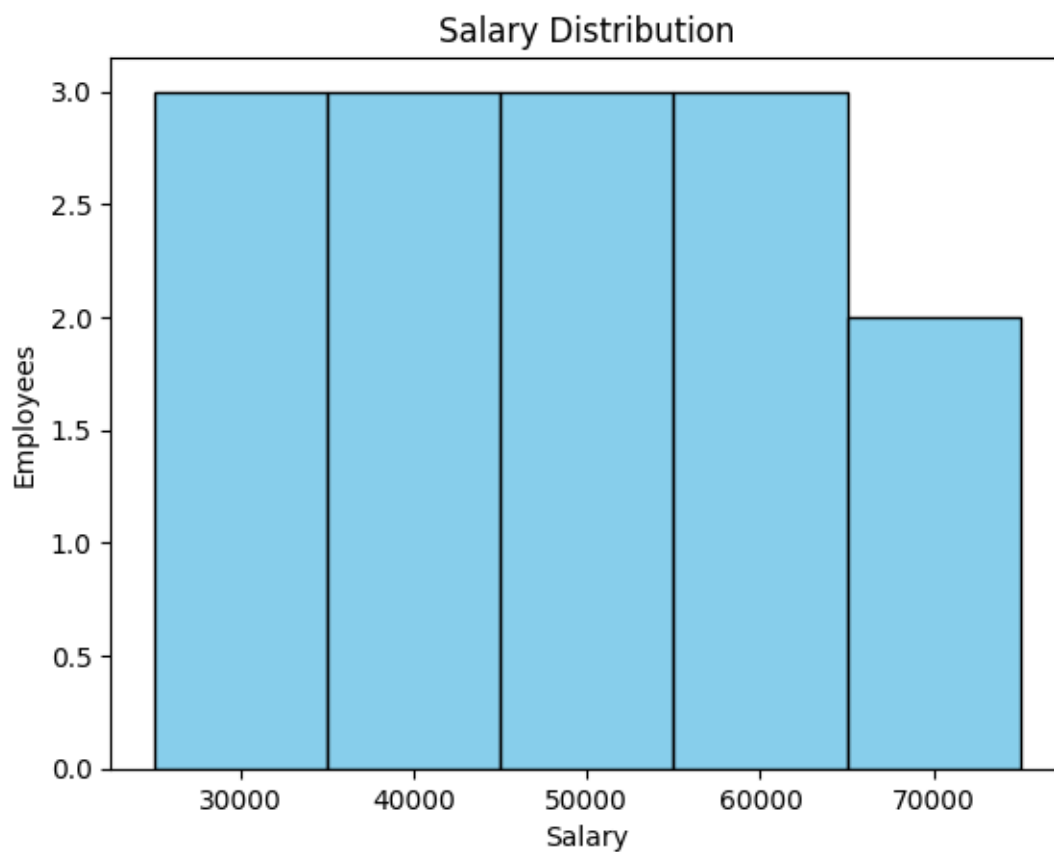
```
import matplotlib.pyplot as plt

salary = [25000, 30000, 32000, 35000, 40000,
          42000, 45000, 48000, 50000, 55000,
          60000, 62000, 70000, 75000]

plt.hist(salary, bins=5, color='skyblue', edgecolor='black')

plt.title("Salary Distribution")
plt.xlabel("Salary")
plt.ylabel("Employees")
plt.show()
```

OUTPUT:



Scenario 7: Smartphone Market Comparison

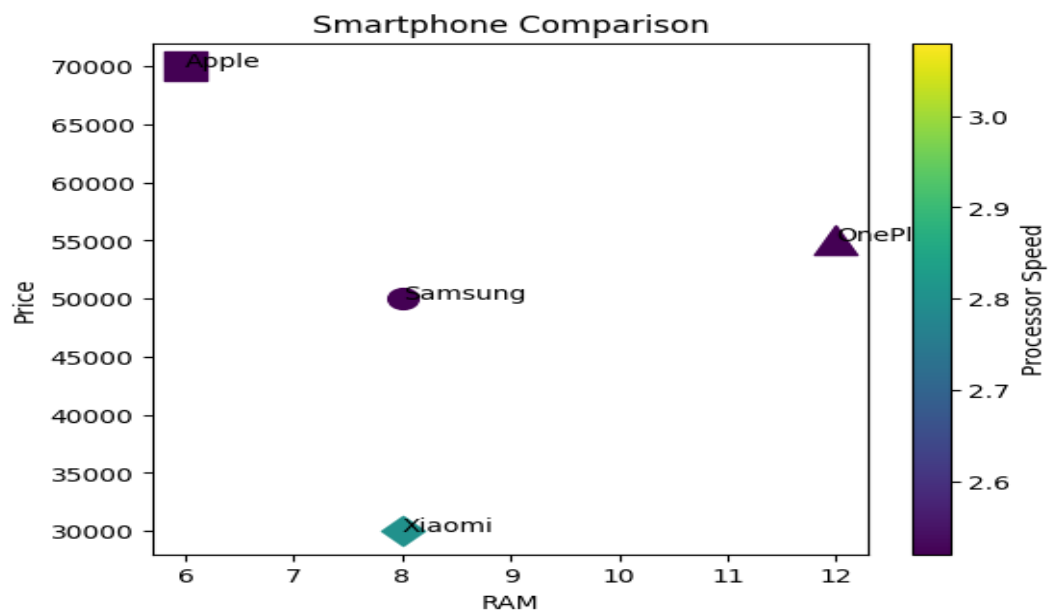
CODE:

```
import pandas as pd
import matplotlib.pyplot as plt

data = {
    'Brand': ['Samsung', 'Apple', 'OnePlus', 'Xiaomi'],
    'RAM': [8, 6, 12, 8],
    'Price': [50000, 70000, 55000, 30000],
    'Storage': [128, 256, 256, 128],
    'Processor': [2.5, 3.2, 3.0, 2.8]
}

df = pd.DataFrame(data)
markers = ['o', 's', '^', 'D']
plt.figure(figsize=(6,5))
for i in range(len(df)):
    plt.scatter(df['RAM'][i],
                df['Price'][i],
                s=df['Storage'][i],
                c=[df['Processor'][i]],
                cmap='viridis',
                marker=markers[i])
plt.text(df['RAM'][i], df['Price'][i], df['Brand'][i])
plt.colorbar(label='Processor Speed')
plt.xlabel("RAM")
plt.ylabel("Price")
plt.title("Smartphone Comparison")
plt.show()
```

OUTPUT:



Scenario 8: Online Movie Rating Analysis

CODE:

```
import pandas as pd
import matplotlib.pyplot as plt

data = {
    'Genre': ['Action', 'Drama', 'Comedy', 'Horror'],
    '2023': [4.1, 4.5, 3.8, 3.9],
    '2024': [4.2, 4.6, 4.0, 3.7],
    '2025': [4.3, 4.8, 4.2, 3.8]
}

df = pd.DataFrame(data)

heat = df.set_index('Genre')

plt.imshow(heat, cmap='YlGnBu', aspect='auto')

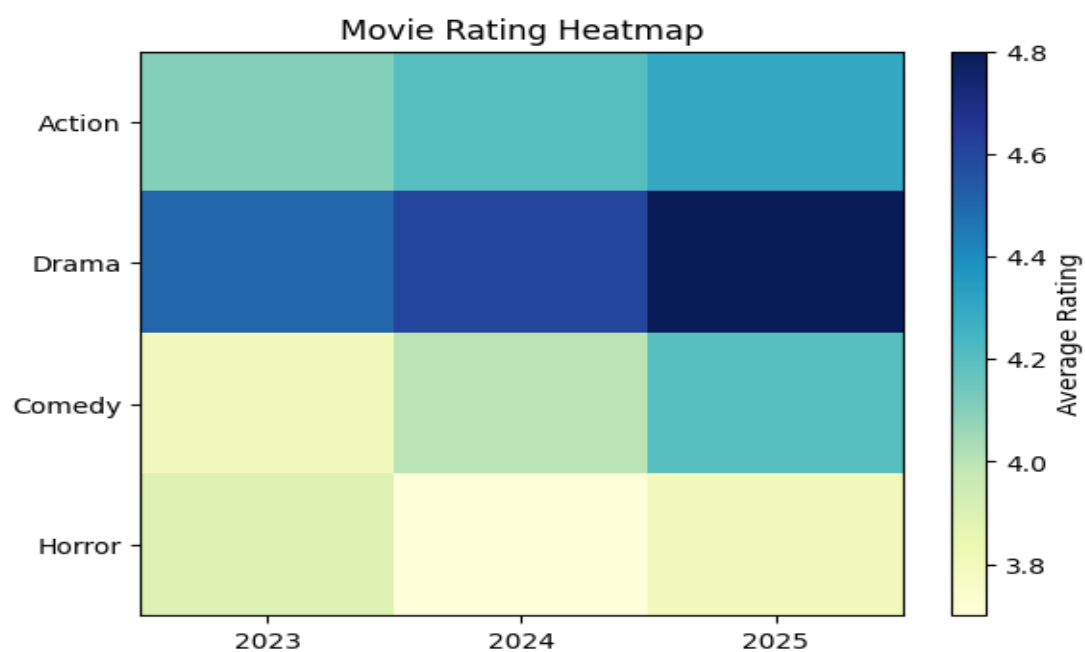
plt.colorbar(label="Average Rating")

plt.xticks(range(3), ['2023', '2024', '2025'])
plt.yticks(range(4), heat.index)

plt.title("Movie Rating Heatmap")

plt.show()
```

OUTPUT:



Scenario 9: COVID-19 State-wise Analysis (Tile Plot)

CODE:

```
import pandas as pd
import matplotlib.pyplot as plt

states = ['TN', 'AP', 'KA', 'KL', 'MH']
cases = [[1200], [1800], [900], [700], [2500]]

plt.imshow(cases, cmap='Reds', aspect='auto')

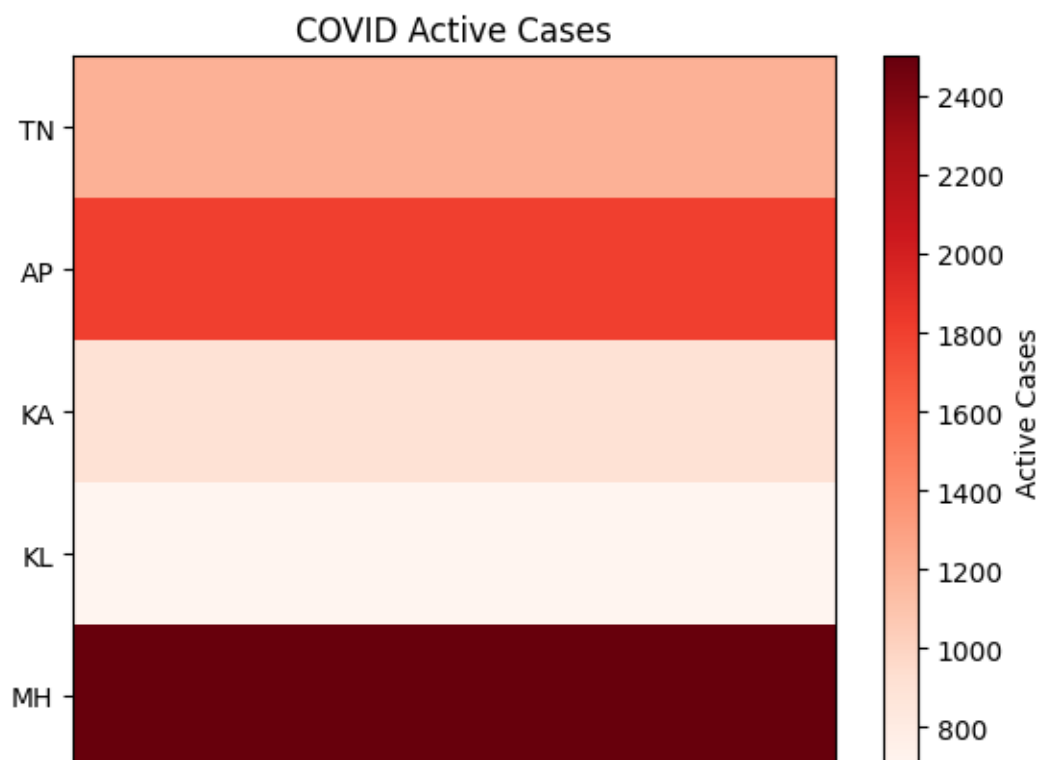
plt.yticks(range(5), states)
plt.xticks([])

plt.colorbar(label='Active Cases')

plt.title("COVID Active Cases")

plt.show()
```

OUTPUT:



Scenario 10: Iris Flower Classification

CODE:

```
from sklearn.datasets import load_iris
import matplotlib.pyplot as plt
iris = load_iris()
X = iris.data
y = iris.target
markers = ['o', 's', '^']
colors = ['red', 'green', 'blue']
plt.figure(figsize=(7,5))
for i in range(3):
    plt.scatter(
        X[y==i,0],      # Sepal Length
        X[y==i,2],      # Petal Length
        s=X[y==i,1]*25, # Sepal Width
        c=colors[i],
        marker=markers[i],
        label=iris.target_names[i]
    )
plt.xlabel("Sepal Length")
plt.ylabel("Petal Length")
plt.title("Iris Flower Classification")
plt.legend()
plt.show()
```

OUTPUT:

